Methods for Estimating Improper Payments in Federal Health Programs: A Narrative Review of Their Comparative Accuracy

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Running title: Comparative accuracy of improper payment estimation methods: a review

Companion study: Cistola A (2026). *Comparative Accuracy of Improper Payment Estimation Methods in Federal Health Programs: A Monte Carlo Evaluation of Analytic Estimators and Bootstrap Simulation* (this repository, `manuscripts/`).

Abstract

Background. The Payment Integrity Information Act of 2019 requires statistically valid improper payment estimates, with confidence intervals, for federal programs at risk of significant improper payments. The Centers for Medicare & Medicaid Services (CMS) produces five such estimates: Medicare Part C, Medicare Part D, Medicaid and CHIP (PERM), the Marketplace advance premium tax credit (EIPM), and Medicare fee-for-service (CERT). Each estimate projects audited errors from a probability sample using either design-based analytic estimators (expansion, ratio, or regression) or simulation that reruns the payment rules.

Objective. The purpose of this review is to summarize the evidence comparing available methods for estimating improper payments in federal health programs. In this review, accuracy has two parts:

- point accuracy;
- interval accuracy (coverage at the nominal level and, after that, width).

Methods. This study is a narrative review of two bodies of work:

- public statute, OMB guidance, and CMS, HHS, GAO, and OIG methodology documents for the five programs and for contract-level RADV audits;
- the statistical, audit-sampling and survey-bootstrap literature on interval accuracy for skewed, zero-inflated payment errors.

Findings. All five programs use stratified probability samples and ratio-type estimators. Part D alone describes its estimate as simulation-based, although the interval algorithm itself is not public, while the others document or imply normal-theory intervals. Interval accuracy depends mainly on two factors (as opposed to the name of the method):

- the skew of the error distribution;
- the number of nonzero errors in the sample.

Normal-theory intervals are conservative at low error rates and liberal at high ones. Second-order bootstrap intervals (BCa and bootstrap-*t*) correct much of the low-error failure. No published study has compared these methods against a known truth in the designs CMS uses (risk-score, drug-benefit, eligibility, APTC, or national claims estimates).

Conclusions. The comparative accuracy of improper payment estimation methods in federal health programs is largely untested, and the methods in production rest on assumptions that the evidence calls into question. This review sets out a research agenda for design-based Monte Carlo validation against synthetic populations, scored on the metric that matches the use of each estimate.

Keywords: improper payments; payment integrity; survey sampling; bootstrap; confidence intervals; Medicare; Medicaid; audit sampling

1. Introduction

Every improper payment estimate in a federal health program does three things:

1. draws a probability sample of payments;
2. establishes the payment error of each sampled unit through review;
3. projects the errors to the program with an uncertainty statement.

These estimates matter. The five CMS measurements reported more than $90 billion in improper payments for FY 2025, and their confidence intervals are audited by Inspectors General. In addition, in some settings the estimates become financial determinations (for example, Medicare Advantage contract recoveries and, beginning in FY 2030, Medicaid federal match disallowances above a 3% eligibility error rate).

The purpose of this review is to summarize the evidence comparing **available methods for estimating improper payments in federal health programs**. This review defines accuracy as in the companion Monte Carlo study:

- **Point accuracy:** how close the estimate is to the true improper payment.
- **Interval accuracy:** whether the 95% interval contains the truth in about 95% of repeated audits (coverage) and, among intervals that do, how narrow the interval is (width).

The review is organized around three questions:

1. **Terminology and practice.** What are the estimation and extrapolation techniques, and how does each program deploy them?
2. **Evidence.** What does the literature say about the accuracy and limitations of each technique?
3. **Fit.** Where would simulation and bootstrap methods help or hurt in each program, and how should their accuracy be tested?

Two inferential targets run through every section and should be kept apart:

- **The national improper payment rate** reported under the Payment Integrity Information Act, which is a program-level ratio with a two-sided confidence interval.
- **The contract- or entity-level recovery amount** (for example, a RADV extrapolation against a Medicare Advantage contract). In this setting, the estimate becomes a legal demand and is usually taken at a lower confidence bound.

A method adequate for the first can still be inadequate for the second, because coverage failures in one tail matter more than overall width.

2. Review approach

This study is a narrative review (as opposed to a systematic review). Two bodies of evidence were searched and read in full where available:

1. **Program sources.** Primary public sources on the design and estimation of each program, including statute and OMB guidance; CMS program pages, error-rate findings, methodology documents, and manuals; OMB information-collection supporting statements; GAO and OIG reports; and court records on RADV extrapolation.
2. **Methods literature.** Statistical, audit-sampling and survey-methods literature on the accuracy of estimators and intervals for skewed and zero-inflated payment errors, including finite-population and survey bootstraps.

Sources were identified by targeted searches, from the references of key papers (especially Edwards, Liu and Delage 2023; Kvanli and Schauer n.d.; Mashreghi, Haziza and Léger 2016; the National Research Council panel of 1988), and from CMS methodology documents. Program facts are taken only from public documents current to October 2026, and anything that would require non-public contractor methodology plans is flagged as

unknown. This review covers the PIIA national rate as its primary target and RADV recovery as context. Ninety-five sources are annotated in the reference list.

3. Regulatory and institutional background

Every estimate in scope exists because the Payment Integrity Information Act of 2019 (PIIA) requires a statistically valid improper payment estimate for programs at risk of significant improper payments, and Appendix C to Office of Management and Budget (OMB) Circular A-123 (M-21-19, March 5, 2021) sets the rules. Payments are "significant" above $10 million and 1.5% of outlays, or above $100 million outright (M-21-19, §II.C).

What Appendix C requires. Each program submits a Sampling and Estimation Methodology Plan (S&EMP) to OMB (Phase 2). OMB reviews the plan but does not formally approve it, and the guidance does not prescribe a confidence level; agencies are to "work with their statistician to determine the appropriate confidence interval." Appendix C also defines the *unknown payment*: one the agency cannot classify as proper or improper because documentation is insufficient. Inspectors General review the plans annually, and interval accuracy is an audited item. For example, the Department of Education Office of Inspector General (OIG) has repeatedly found estimates unreliable because nonrandom student-level sampling made the confidence intervals inaccurate, a finding first reported for FY 2019 and repeated through the FY 2025 review (ED-OIG/A26DC0288, August 4, 2026). This precedent matters here because a change in interval method is something an OIG can test.

Precision targets. Medicare FFS (CERT) designs for at least a 95% confidence interval of ±3 percentage points (CMS CERT overview). PERM's older statutory target was a 90% interval of ±2.5 points nationally, and CMS adds its own state-level goal of 95% ±3 points (MACPAC).

Who does the work. The Payment Accuracy and Reporting Group (PARG) in CMS's Office of Financial Management runs improper payment measurement for Medicare FFS, Parts C and D, Medicaid, CHIP, and APTC, across 11 contracts and about $114 million a year (PARG director bio; no primary CMS budget document states these figures). One boundary is important to note. The national Part C measure is a PARG product, but *contract-specific* RADV recovery audits are planned and extrapolated by CMS's Center for Program Integrity (PY 2019 RADV methods). The two share review machinery but not inferential goals.

FY 2025 results (HHS Agency Financial Report, November 2025; CMS fact sheet, January 15, 2026):

Program	Measure	FY 2025 rate	Improper $	Data year
Medicare Part C	Part C IPM	6.09%	$23.67B	CY 2023
Medicare Part D	Part D IPM	4.00% (95% CI 3.57–4.46%)	$4.23B	CY 2023
Medicaid	PERM	6.12%	$37.39B	rolling 3 cycles
CHIP	PERM	7.05%	$1.37B	rolling 3 cycles
Marketplace APTC (FFE only)	EIPM	0.89%	$657.46M	CY 2023
Medicare FFS	CERT	6.55% (95% CI 5.9–7.2%)	$28.83B	Jul 2023–Jun 2024 claims

HHS was not fully PIIA-compliant for FY 2025 (HHS OIG, OAS-26-17-042; PaymentAccuracy.gov). State-based Exchange APTC is not yet measured; the State Exchange improper payment measurement rule applies beginning January 1, 2027 (45 CFR §155.1605).

4. Terminology of error estimation and extrapolation

Every measure in scope does the same three things: draw a probability sample, establish a payment error for each sampled unit through review, and then project ("extrapolate") to

the frame and attach an uncertainty statement. The programs differ in the estimator, the variance method, and what the interval is *used for*. These three choices determine which method is most accurate in each setting.

Sampling and review vocabulary

Term	Meaning	Where it appears
Universe / target population	All payments the estimate is meant to describe	PERM universe of federally matched payments; Part C payments subject to risk adjustment
Sampling frame	The list sampled from; an extrapolation describes the frame only	RADV PY 2019+ frames restricted to enrollees a prediction model ranks as high-risk
Sampling unit	What is drawn: claim, PDE, beneficiary, application, enrollee	Enrollee (Part C, RADV); beneficiary (Part D since FY 2025); application (EIPM); payment (PERM)
Stratum / certainty stratum	Disjoint subgroups sampled separately; a take-all stratum reviews every unit	Risk-score terciles (Part C, 2012 RADV); payment strata within quarter (PERM)
Payment error y	Paid minus amount supported by documentation; positive = over, negative = under	Every program
Gross vs net error	Gross sums absolute over- and underpayments; net nets them	PIIA reporting uses gross; recovery uses net overpayment
Unknown / insufficient documentation	Payment that cannot be classified; CMS programs count it as improper	75% of Part D sampled errors in FY 2025; a subset of Part C overpayments
Taint	Error as a share of the payment, y/x	Audit literature; MUS bounds
All-or-nothing error	Unit is either fully in error or fully correct	Common in Medicare FFS investigations; missing-record errors in all programs

Estimators

The **expansion (mean-per-unit, Horvitz–Thompson) estimator** weights each sampled error by the inverse of its selection probability. This is the estimator used by 2012 RADV and PIM Chapter 8 contractors, and it is the default in the HHS OIG's RAT-STATS software, which is the de facto standard implementation of normal-theory bounds in Medicare integrity audits (HHS OIG, RAT-STATS). Standard textbook treatments are Cochran (1977) and Lohr (2022).

$$\hat{T} = \sum_{h=1}^{L} \frac{N_h}{n_h} \sum_{i \in s_h} y_{hi}, \qquad \hat{V}(\hat{T}) = \sum_{h=1}^{L} N_h^2 \left(1 - \frac{n_h}{N_h}\right) \frac{s_h^2}{n_h}$$

The **ratio estimator** divides projected improper dollars by projected paid dollars. Improper payment *rates* (PERM, Part C, Part D, APTC) are ratios, so their variance comes from linearization or replication (as opposed to the simple formula above). Current RADV (PY 2019 onward) estimates the mean change in risk score and then multiplies it by the summed county rates of the frame, which is effectively a ratio-type estimator.

$$\hat{R} = \frac{\sum_{h} \frac{N_h}{n_h} \sum_{i \in s_h} y_{hi}}{\sum_{h} \frac{N_h}{n_h} \sum_{i \in s_h} x_{hi}}$$

Difference and regression estimators use the known paid amount x as an auxiliary variable. These estimators are standard in audit sampling (Kaplan 1973 was the first to show that the zero/nonzero mixture of audit populations undermines their normal-theory intervals), but are not documented in the public PARG methods. The choice between stratified expansion and ratio estimation in Medicare and Medicaid benefit integrity investigations is treated directly by Edwards (2011), which appears to be the only published treatment of that choice for these audits. Valliant, Dorfman and Royall (2000) give the prediction (model-based) view of these estimators, which is the frame in which the auxiliary information of the full population can be used; this view also describes what the simulation methods below do. **Composite estimators** combine separately estimated components. For example, the PERM overall rate is claims rate plus eligibility rate minus

an overlap term, and the national rate rolls three state cycles together.

Variance and interval methods

Method	How the interval is formed	Known weakness in audit data
Normal / *t* with analytic SE	Estimate ± critical value × SE; *z* or *t* with $n-1$, $n-L$ or Satterthwaite df	Assumes near-symmetric sampling distribution; fails under strong skew, many zeros, small *n*
Taylor linearization	Delta-method SE for ratios, then normal interval	Same normality assumption; unstable when the denominator is small or variable
Jackknife (JK1, JKn), BRR	Replicate weights drop or reweight PSUs (Krewski and Rao 1981; Wolter 2007)	BRR needs 2 PSUs per stratum; jackknife is poor for non-smooth statistics
Naive iid bootstrap	Resample *n* units with replacement, recompute	Ignores FPC and strata; overstates variance when sampling fraction is large
Survey bootstraps (McCarthy–Snowden, Rao–Wu rescaling, mirror-match, pseudo-population, generalized)	Resample within strata with rescaled weights to reproduce design variance (Bickel and Freedman 1984; Chao and Lo 1985; Rao and Wu 1988; Sitter 1992a, 1992b; Beaumont and Patak 2012)	Need ≥2 units per stratum; odd behavior with tiny strata
Percentile / basic interval	Quantiles of the bootstrap distribution	Too narrow for small *n*; inherits skew but not bias
Bootstrap-*t*	Studentize each replicate, invert its quantiles	Sensitive to unstable SE estimates in replicates
BCa	Percentiles shifted for bias (z_0) and skewness (acceleration *a*)	Best second-order accuracy in theory; acceleration noisy at very small *n*
Empirical likelihood	Profile likelihood ratio under a nonparametric model	Promising for zero-inflated means; until recently untested at high error rates
MUS / Stringer and hypergeometric bounds	Bounds built from the count and size of errors (Stringer 1963; Fienberg, Neter and Leitch 1977; Bickel 1992)	Conservative by design; minimal recovery when errors are frequent

The BCa lower limit replaces the nominal α with an adjusted level read off the bootstrap distribution:

$$\alpha_{\mathrm{BCa}} = \Phi\left(\hat{z}_0 + \frac{\hat{z}_0 + z_\alpha}{1 - \hat{a}\,(\hat{z}_0 + z_\alpha)}\right), \qquad \mathrm{LB} = \hat{\theta}^*_{(\alpha_{\mathrm{BCa}})}$$

Simulation-based estimation

A third family of methods simulates (as opposed to projects). In **audit-finding simulation**, the audit findings are assigned at random to the unaudited units of the whole population, the program's payment function is rerun, and the process is repeated many times. The findings are which HCCs failed, which drug claims were wrong and by how much, and which applications were ineligible. The estimate and interval are read from the distribution of simulated improper payments.

This method is attractive for three reasons:

- **Nonlinear payments.** The simulation reruns the actual payment rules (for example, Part D benefit phases, risk-score hierarchies, and the APTC schedule).
- **Every measure at once.** One set of iterations produces over-, under-, gross, and net payments, along with any subgroup.
- **Traceability.** Every simulated dollar can be traced to an audited finding.

This method differs from the survey bootstraps in the table above in one important way: the audit itself is not resampled. Two versions are possible, and this review names them differently because only one of them is a bootstrap in the usual sense:

- **Plug-in audit-finding simulation (fixed rates).** The error rates estimated from the audit are reused in every iteration. Nothing that was observed is resampled, so the method is a plug-in microsimulation (a stochastic reconstruction of the population under point-estimated rates) and not a bootstrap. Its spread measures only the variability of reassigning findings across the population.
- **Parametric-bootstrap audit-finding simulation (parametric rates).** The rates

are redrawn from their sampling distribution (for example, binomial) before each iteration, which represents the uncertainty of the audit itself. This is a parametric bootstrap of the rates (Efron and Tibshirani 1993, ch. 6).

The distinction is the same one that separates single from multiple imputation. A single imputation treats the imputation model as known and understates variance by the between-imputation component; multiple imputation redraws the model parameters and recovers it (Rubin 1987). By the law of total variance, the plug-in version omits the between-audit component in the same way, so its interval is expected to be too narrow by an amount that grows as the population grows relative to the audit. The parametric version is a frequentist device, as opposed to the Bayesian bootstrap of Rubin (1981), which places a posterior on the observed values rather than on a parametric rate. Whether the interval from either version holds its nominal coverage is an empirical question addressed in Sections 6 and 8.

Decision quantities and performance language

- **Lower confidence bound (LB).** PIM Chapter 8 recovers the lower limit of a *one-sided* 90% interval; the 2012 RADV method used the lower limit of a *two-sided* 99% interval (one-sided 99.5%); PY 2019–2021 RADV use the lower limit of a 90% interval only as a gate and recover the point estimate. The PY 2019 bound uses $z = 1.645$, which is the lower limit of a two-sided 90% interval and therefore a one-sided 95% bound; the PY 2021 document specifies $t(n-1)$ without stating whether the 90% is one- or two-sided. The difference between the two readings is a frequent source of confusion in appeals. Gilliland and Edwards (2011) propose randomized confidence limits as one way to balance the risk between the government and the provider in this setting.
- **Coverage and one-sided non-coverage.** The share of repeated samples in which the interval (or bound) contains the true total. For recovery, the operative metric is how often LB exceeds the truth, nominally 10% for a one-sided 90% bound.
- **Viability and recovery percent.** Edwards et al. call a design “viable” if it achieves at least 89% coverage at every tested error rate, and measure efficiency as average recovery as a share of the true overpayment.
- **Monte Carlo vs bootstrap.** *Monte Carlo* here means simulating from a model or a synthetic population (used to validate designs and to propagate component uncertainty). *Bootstrap* means resampling the observed sample, or redrawing a parameter from its estimated sampling distribution. CMS’s Part D documents use “simulation” for both the benefit re-adjudication and the interval iterations, so the word alone does not indicate which is meant. A Monte Carlo study of an interval method is itself a statistical experiment, and its coverage estimates carry Monte Carlo error that should be reported with them (Koehler, Brown and Haneuse 2009; Morris, White and Crowther 2019).

5. How each program measures improper payments

Only one of the five PIIA measures (Part D) publicly describes its estimate as simulation-based, while the others document (or imply) analytic normal-theory intervals. RADV is the outlier in terms of stakes, with small samples, a one-sided decision rule, and money changing hands on the result.

Program	Sampling unit and design	Sample size	Estimator	Interval (public record)	What the interval decides
Part C IPM (PARG)	Enrollee; stratified random sample, historically high/medium/low risk-score strata	Not stated in public results	Enrollee payment error projected to Part C payments subject to risk adjustment	Not stated in the FY 2025 results document	PIIA rate only
RADV, 2012 notice (CPI)	Enrollee; three equal risk-score terciles	201 (67 per stratum)	Stratified expansion	Two-sided 99%, $z = 2.575$	Recover the lower bound, less an FFS adjuster
RADV, PY 2019 (CPI)	Enrollee; SRS from a frame of the top	35	Mean change in risk score × Σ	Lower limit of a two-sided 90%	Recover point estimate if the

	decile by CPI's improper-payment prediction models		county rates	interval (one-sided 95% bound), z = 1.645, with FPC	bound is above 0
RADV, PY 2021 (CPI)	Same, frame widened to the top quartile	35, 50, 100 or 200 by contract-size stratum	Same; Taylor linearization for partial-year enrollees	Lower bound of a 90% interval, t with $n-1$ df (sidedness not stated)	Extrapolation designed; collection currently sample-only
Part D IPM (PARG)	Beneficiary (random selection since FY 2025; cohort before); PDEs validated against prescriptions	Audit sample not stated publicly; findings extrapolated through a random 5% sample of Part D beneficiaries	Corrected PDEs re-run through a benefit simulation to get payment error	95% interval; estimate described as simulation-based, algorithm not public	PIIA rate only
PERM (PARG)	Two phases: states rotate in 3 cycles; within state, quarterly stratified samples of FFS claims and MC payments; eligibility cases attached to sampled payments	Set per state from expenditures and prior error; national target 95% ±3 pts	Stratified ratio (projected improper ÷ projected paid); claims + eligibility − overlap; 3-cycle rolling national rate	Analytic stratified ratio variance per the OMB supporting statement (not independently verified)	PIIA rate; state CAPs; from FY 2030, disallowance above 3% eligibility error
EIPM, APTC (PARG)	FFE application; stratified random sample (strata not public)	~2,000	Projected APTC in error ÷ APTC paid	95%; margin of error ±0.54 points (FY 2025)	PIIA rate only
CERT, Medicare FFS (PARG)	Claim; stratified random sample by claim type (Part A IPPS, Part A excluding IPPS, Part B, DMEPOS), then about 100 service-level strata per type (type-of-bill strata for Part A excluding IPPS)	About 37,500–50,000 reviewed (37,651 in FY 2025)	Weighted projection; reduces nationally to the separate ratio estimator; contractor clusters weighted by expenditure share	95%, designed for ±3 points	PIIA rate; MAC recoupment of sampled overpayments

Medicare Part C: national measure and contract audits

The Part C IPM verifies that HCCs submitted for payment are supported in the medical record for a sampled enrollee, computes the enrollee-level payment error, and projects this error to the Part C population subject to risk adjustment (CMS Part C IPM). The FY 2025 rate was 6.09% ($23.67B gross, $19.20B net) on a $388.72B denominator, and missing documentation was only 0.22% of expenditures (FY 2025 results). One definitional change affected the series: "spontaneous" additional HCCs were removed from the underpayment calculation beginning in CY 2020. The results document does not state the sample size, the strata or the interval method.

Contract-specific RADV has gone through four statistical regimes in fourteen years:

Payment year (document date)	Frame	*n*	Interval	Recovery rule
PY 2021 (May 29, 2026)	Top quartile by prediction model	35–200 by contract size	Lower bound of a 90% interval, $t(n-1)$; sidedness not stated	Collect sampled enrollees only; extrapolation reserved
PY 2019 (June 25, 2025, v2)	Top decile by prediction model	35	Lower limit of a two-sided 90% interval (one-sided 95%), z = 1.645	Point estimate if bound > 0
PY 2018 (2025, per Marshall)	Prediction models plus chart-review contracts	Not stated	90%	Lower bound
PY 2011+ (Feb 24, 2012)	All RADV-eligible enrollees	201	99% two-sided, z	Lower bound less FFS adjuster

The legal basis for these audits is unsettled. The 2023 RADV final rule (88 FR 6643; no FFS adjuster; extrapolation from PY 2018) was vacated on procedural grounds under the Administrative Procedure Act in *Humana Inc. v. Becerra*, No. 4:23-cv-00909-O (N.D. Tex. Sept. 25, 2025). The government appealed as *Humana Inc. v. Kennedy*, No. 25-11293 (5th Cir.), and a Fifth Circuit panel heard argument on October 5, 2026 (STAT). Meanwhile,

CMS plans to audit all eligible contracts (approximately 550) each year, with 35 to 200 records per contract (CMS press release, May 21, 2025; RADV fast facts). Regardless of the court's decision, the estimator and bound will likely be argued over contract by contract, which is exactly where coverage evidence matters.

Medicare Part D

The Part D IPM samples beneficiaries, requests documentation for their PDEs, corrects gross drug cost where the record does not support the PDE, and runs the corrected PDEs through a simulation of the benefit year so that errors flow through benefit phases and reinsurance (CMS Part D IPM; HHS performance plan). The program's background page states that the validation findings "are applied to a randomly selected 5% sample of the Part D beneficiary population," that the payment error is calculated for that sample, and that the result is extrapolated to the remaining beneficiaries (Part D IPM program background). This 5% sample is the extrapolation base (as opposed to the audited sample), and it is the closest public description of an audit-finding simulation in any of the five programs. The FY 2023 redesign changed the sampling unit, reduced "the number of simulation iterations to align to statistical literature," and aligned the confidence interval to the estimation method (FY 2023 results). In FY 2025, the program moved from cohort to random beneficiary selection.

The published intervals are slightly asymmetric in FY 2023 ($2.63B–$4.17B around $3.35B, or −0.72/+0.82) and nearly symmetric in FY 2025 ($3.77B–$4.71B around $4.23B, or −0.46/+0.48) (FY 2025 results), so the published intervals do not by themselves reveal the algorithm. Neither findings document states that the interval is read from the simulation iterations. Public documents do not name the algorithm, so the internal S&EMP is the authority on this point. Missing or invalid documentation was 75% of sampled errors in FY 2025, which makes the error distribution closer to all-or-nothing than to small pricing discrepancies.

Medicaid and CHIP (PERM)

PERM measures each state once every three years and builds a national rolling rate from the latest three cycles (PERM Manual, April 2025). Within a state, quarterly claims universes are stratified by payment (five payment strata for managed care in recent cycles; RY 2022 kickoff) and, per the OMB supporting statement (OMB control no. 0938-0974; ICR 202408-0938-026; not independently verified for this review), selected systematically within payment strata after sorting by paid amount. Preston (2009) gives a rescaled bootstrap for stratified multistage designs of this kind. Each component rate is projected improper dollars over projected paid dollars, and the overall rate adds the eligibility rate to the claims rate and subtracts their overlap (CMS rate calculation).

The stakes for state-level estimates changed in 2025. P.L. 119-21 §71106 (July 4, 2025) disallows federal match on eligibility improper payments above a 3% rate starting October 1, 2029, and removes the good-faith waiver at 42 CFR 431.1010 (statute; summarized by BPC 2026). A state eligibility rate near 3% now carries a dollar consequence, so its interval behaves like a recovery bound (as opposed to only a reporting statistic). RY 2027 measures eligibility only (RY 2027 kickoff).

Marketplace APTC (EIPM)

EIPM reviews a stratified random sample of about 2,000 FFE applications to test whether eligibility determinations and APTC amounts were correct (EIPM background; CMS 2022 release). Rates have stayed near or below 1% (0.62% in the first FY 2022 report; 0.58% in FY 2023; 1.01% in FY 2024; 0.89% in FY 2025, margin of error ±0.54 points). No underpayments were reported for FY 2023–2025 (PaymentAccuracy.gov). In FY 2025, 0.43 of the 0.89 points were *technically improper* payments, where a required periodic data match was not run (CMS EIPM improper payment findings). The remainder came from manual errors (for example, income entered incorrectly or documents accepted despite name or date-of-birth discrepancies). At this rate, a 2,000-unit sample contains only a few dozen errors, which is the rare-error setting of Cox and Snell (1979) and the setting in

which normal-theory bounds are least reliable. State-based Exchanges come under a separate SEIPM rule (45 CFR 155 Subpart Q) with state-specific sample sizes and a three-year cycle, applicable beginning January 1, 2027 (45 CFR §155.1605).

Medicare fee-for-service (CERT)

CERT reviews a stratified random sample of Medicare FFS claims submitted from July through June, including both paid and denied claims (CERT introduction).

- **Strata.** The first level is claim type (Part A hospital IPPS, Part A excluding IPPS, Part B, and DMEPOS). Within each type, there are about 100 service-level strata, and Part A excluding IPPS uses two-digit type-of-bill strata, fewer than 20 (CERT methodology).
- **Estimator.** A generalized estimator produces national, contractor-cluster, and service-level rates. At the national level it reduces to the separate ratio estimator, with contractor clusters weighted by their share of expenditures.
- **FY 2025 sample.** 49,061 claims were sampled and 37,651 reviewed: IPPS 8,750; Part A excluding IPPS 8,651; Part B 11,600; DMEPOS 8,650.
- **FY 2025 results.** The gross rate was 6.55% (95% CI 5.9–7.2%), with $27.9B in overpayments and $1.0B in underpayments, counted gross (2025 Supplemental Improper Payment Data). By type, the rates ranged from 3.1% (IPPS) to 24.1% (DMEPOS).
- **Error categories.** Insufficient documentation accounted for 53% of improper dollars, medical necessity 15%, no documentation 12%, incorrect coding 11%, and other 8.5%.

Most CERT errors are therefore whole-claim, all-or-nothing errors, with partial errors coming mainly from coding. This is the error setting studied by Edwards, Liu and Delage (2023), but at a national sample size where the central limit theorem has far more to work with than in a 30-claim provider audit.

Unknown from the public record:

- the Part C IPM sample size, strata, and interval method;
- Part D's resampling algorithm and number of replicates;
- the PERM national variance formula;
- EIPM's strata;
- CERT's allocation rule and variance method (described in an appendix to the Supplemental Improper Payment Data report).

6. Evidence on comparative accuracy

The literature converges on one rule: whether an interval holds its nominal coverage depends mainly on the skew of the error distribution and the number of nonzero errors in the sample (as opposed to the name of the method). Bootstrap intervals correct the low-error, right-skew failure well. However, they have barely been tested in the high-error, all-or-nothing setting that Medicare reviews actually produce.

Normal-theory intervals

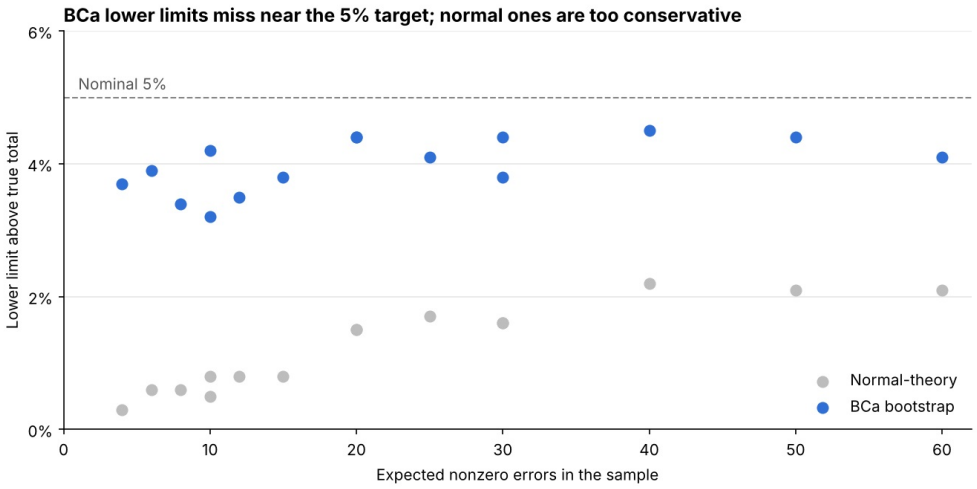
- **The principle of skew.** A right-skewed population produces a conservative lower bound and a liberal upper bound, and a left-skewed population reverses this pattern (Cochran 1977, pp. 39–44, as applied by Edwards, Liu & Delage 2023). Overpayment populations are a spike at zero plus a payment-shaped tail, so their skew reverses as the error rate rises.
- **Low error rates: too conservative.** With few nonzero errors, normal lower limits sit far below the truth. In Kvanli and Schauer's simulations, the traditional 90% interval's lower limit exceeded the true total in only 0.3–2.3% of samples as opposed to a nominal 5% (Kvanli and Schauer n.d.). Neter and Loebbecke's 1975 AICPA study of real accounting populations is the classic early demonstration that standard estimators' intervals miss nominal coverage when errors are rare, and Kaplan (1973)

had already identified the mixture structure as the cause. Cox and Snell (1979) give the canonical model for sampling and estimating rare errors, and Rohrbach (1993) proposes augmenting the variance estimate so that the interval reaches nominal coverage in audit populations, which is a direct precursor of the coverage problem this review addresses.

- **High error rates: too liberal.** In 166 Medicare integrity samples, more than 76% of payments were in error and the data fit an all-or-nothing model. Under this model, a simple random sample of 30 fell below 90% confidence once the error rate passed 0.6 for two of three test populations, and no stratification algorithm except the geometric (Geo) algorithm of Gunning and Horgan (2004) was viable for more than about half of the 104 test populations; the geometric algorithm reached 88% in one design cell (Edwards et al. 2023). At very high error rates, the bound can even exceed total payments.
- **Stratification can make it worse.** Payment-based strata can create left-skewed overpayment strata at high error rates, and King and Madansky (2013) warn that careless stratification can invalidate the confidence coefficient itself. The stratification of skewed payment populations is reviewed by Horgan (2006).
- **CMS acknowledges the gap.** PIM Chapter 8 accepts “standard methods” from texts and journals but acknowledges that 90% coverage cannot be guaranteed in every audit (PIM Ch. 8, §8.4.5).

Bootstrap intervals

- **Theory.** Percentile intervals are first-order accurate, while BCa and bootstrap-t are second-order accurate, correcting for skewness at rate $1/n$ rather than $1/\sqrt{n}$ (Hall 1988; Efron 1987; DiCiccio & Efron 1996). In small samples, the percentile interval is too narrow, and bootstrap-t handles skew better (Hesterberg 2015).
- **Audit populations with rare errors.** Kvanli and Schauer found BCa lower limits exceeded the truth in 3.2–4.5% of samples (nominal 5%), much closer than the traditional method, with average lower limits 5.5–531% higher depending on the expected number of errors. The gain fades after about 30 *expected* errors in the sample, at which point both methods agree; the authors conclude that the expected number of errors, as opposed to a minimum count of nonzero errors actually observed, is what drives the gain. On twelve real tax-audit samples, BCa lower limits were 3–548% higher. Earlier bootstrap bounds for audit populations are Biddle, Bruton and Siegel (1990), who bootstrapped the difference and ratio estimators, and Howard (1994), who combined Hoeffding’s inequality with the bootstrap to obtain a bound for overstatement error.
- **Survey designs.** Rescaled bootstraps reproduce design variance under stratified sampling (Rao & Wu 1988; Rao, Wu & Yue 1992), building on the first bootstrap for stratified samples by Bickel and Freedman (1984). Simulation comparisons ranked coverage bootstrap > BRR > jackknife > linearization, with the bootstrap best for one-sided intervals (Krewski & Rao 1981; Rao & Wu 1988; Kovar, Rao & Wu 1988; together with a 1996 review by Shao that was not consulted directly, as summarized collectively by Kolenikov 2008). Sitter (1992b) compares three survey bootstraps and finds none dominant. A naive iid bootstrap ignores the finite population correction and overstates variance (Mashreghi, Haziza & Léger 2016).
- **Rare-error refinements.** Helmers (2000) combines a Cornish–Fisher expansion with bootstrap calibration to get upper limits for total error when errors are rare and highly skewed (Biometrika). Berger, Chiodini and Zenga (2021) give an adjusted empirical likelihood bound for monetary-unit sampling that is designed for the same regime.
- **Skewed health spending.** The cost-effectiveness literature adopted the nonparametric bootstrap for skewed per-person costs (Barber & Thompson 2000), a close analogue to per-enrollee payment error.



Source: Kvanli & Schauer (Multistate Tax Commission), Tables 1–3; 15 population–sample-size cells, minimum 3 nonzero errors per sample.

Share of simulated samples whose 90% lower limit exceeded the true total, normal-theory vs BCa bootstrap

Source: Kvanli & Schauer (n.d., unpublished manuscript hosted by the Multistate Tax Commission), Tables 1–3 · 15 population–sample-size cells, minimum 3 nonzero errors per sample.

Even at 60 expected errors, the normal-theory bound misses less than half as often as it should, so it under-recovers. In contrast, BCa remains slightly conservative throughout. Populations had 2%, 5%, and 10% error rates with exponential error sizes and 5,000 samples per cell.

Where the bootstrap is expected to fail or is untested:

1. **Few or no nonzero errors.** If the sample has zero or one error, every resample is degenerate and the bound collapses to zero or to the single error. EIPM-scale error rates and small RADV contracts are at risk.
2. **Bounded, all-or-nothing errors at high rates.** Resampling cannot produce values outside the observed support and inherits the skew of the sample. Whether this corrects the normal bound's liberal behavior at error rates above 0.6 has not been published for Medicare data.
3. **Tiny strata and systematic samples.** Rescaling requires at least two units per stratum, and the sorted systematic selection used by PERM has no exact design-based variance, so any bootstrap treats it as implicitly stratified.
4. **Model-ranked frames.** RADV PY 2019+ frames are chosen by a prediction model, so inference applies only to the frame. A bootstrap inside the frame is valid, but extrapolating its interval beyond the frame is not.
5. **Composite and multi-phase estimators.** Rolling PERM rates and claims-plus-eligibility composites require resampling each phase and component on its own design, and then recombining them. Correlations between components (the overlap term) must be carried through.
6. **Reproducibility.** Tail quantiles need thousands of replicates, and appeals require that the result be replicable, so the seed, replicate count, and Monte Carlo error must be documented as PIM §8.4.4.2 already requires for sample selection.

Alternatives with published evidence

Approach	Evidence	Limitation	Sources
Empirical likelihood	Better-balanced tails than normal-theory for zero-heavy means at low error rates	Not tested at high error rates until Liu's 2025 work on Medicare populations (preliminary results)	Chen, Chen & Rao 2003; Berger, Chiodini & Zenga 2021; Liu 2025 (ISI WSC)
Likelihood methods for zero-inflated means	Standard *t*-intervals unreliable for many-zero populations	Parametric assumptions on the nonzero part	Kvanli, Shen & Deng 1998; Zhou & Tu 2000; Tian 2005
Hypergeometric / minimum-sum / penny sampling	Guaranteed at least nominal coverage if negative errors	Conservative; lower recovery	Edwards et al. 2003; Ignatova & Edwards 2008; Gilliland &

	are rare; never exceeds total payment		Feng 2010; Edwards et al. 2015
MUS / Stringer-type and multinomial bounds	Standard in financial audit for low-error overstatement	Conservative; ill-suited to frequent or two-sided errors	Stringer 1963; Fienberg, Neter & Leitch 1977; NRC Panel 1988; Bickel 1992
Model-based design testing	Monte Carlo under an all-or-nothing model reveals viability before sampling	Model must match the error process	Edwards et al. 2023 (the authors' `samptest` R function, available on request; stratification algorithms from the CRAN package `stratification`)

The gap this review identifies. No published study compares the accuracy of the available methods against a known truth in the settings CMS measures:

- RADV-style risk-score extrapolation;
- Part D PDE-to-payment simulation;
- PERM eligibility rates;
- APTC errors;
- the national CERT claims estimate.

This gap applies to analytic, bootstrap, and audit-finding simulation methods alike. The closest evidence comes from Medicare FFS provider audits (high error, all-or-nothing) and tax and accounting audits (low error, right skew). The CMS programs span both settings. In addition, before the companion study the validity of audit-finding simulation intervals, whether plug-in or parametric-bootstrap, had not been tested. The companion study's first-stage results on this last point are summarized in Section 8.

7. Fit of simulation and bootstrap methods by program

The case for the bootstrap is strongest where samples are small, errors are skewed, and an interval drives a dollar decision (RADV first, then PERM state eligibility rates, and then EIPM). Part D already relies on simulation, so the question for Part D is validation (as opposed to adoption). Part C IPM gains the least.

Program	Error regime	What the interval decides	Bootstrap priority	Variant to test first
RADV contract audits	n = 35–200 per contract; high error in a model-targeted frame; discrete HCC-sized jumps; some offsets	Gate for extrapolation; historically the recovery amount	High	Without-replacement (pseudo-population) bootstrap-t and BCa on the enrollee-month ratio
PERM, state eligibility	Few hundred cases per state; rates around 1–12%; mostly documentation errors	From FY 2030, federal match disallowed above 3%	High	Rescaled bootstrap within payment strata, propagated through the composite
EIPM (APTC)	~2,000 applications; rate under 1%; few dozen errors; right-skewed	PIIA rate; SBE state rates from 2027	Medium–high	Stratified rescaled bootstrap with BCa; empirical likelihood as comparator
Part D IPM	Beneficiary clusters of PDEs; 75% of errors are missing documentation; nonlinear benefit simulation	PIIA rate	Validate existing	Audit the current resampling scheme's coverage under a synthetic population
Part C IPM	Large national stratified sample; ratio estimator	PIIA rate	Low	Rao–Wu–Yue replicate weights as a check on linearization
CERT	National sample of about 37,500 claims in four claim types; mostly all-or-nothing errors; DMEPOS rate above 20%	PIIA rate (national and by claim type)	Low–medium	Stratified design bootstrap by claim type; check high-error strata (DMEPOS) separately

RADV contract audits

Benefits.

- At $n = 35$ the normal or t bound rests entirely on the CLT for a skewed, discrete variable (change in risk score). Bootstrap-t and BCa correct the skewness term the t bound ignores.
- The PY 2021 estimator is a ratio over enrollee-months. Resampling enrollees (with all their months and HCCs) produces its variance without the linearization step.
- Resampling can bound the dollar payment error directly, including county-rate weighting (as opposed to bounding the mean risk-score change and multiplying afterward).

Drawbacks.

- The decision rule is one-sided: extrapolate if the bound exceeds zero, and then recover the point estimate. What matters is the false-extrapolation rate when the true mean change is at or below zero. At $n = 35$, percentile intervals are known to be too narrow, which would *raise* this rate.
- If the targeted frame yields mostly right-skewed changes (many zeros, some large HCC deletions), a better-calibrated bound can be higher than the t bound, as in Kvanli and Schauer. This would shift recoveries upward and would likely invite challenge.
- Inference applies only to the prediction-model frame, and the bootstrap cannot resolve the frame definition or the FFS-adjuster dispute.
- A method change during litigation is itself a legal event, so any bootstrap must be fully replicable from a documented seed.

Implementation. Each contract should be treated as SRS without replacement from its frame, using a pseudo-population or rescaled bootstrap so that the FPC is honored when n/N is large (frames can be as small as 30). Enrollees should be resampled as clusters, the month-weighted mean change and the dollar estimate computed in each replicate, and bootstrap-t and BCa bounds formed with at least 5,000 replicates. The probability that the bound exceeds zero should be reported alongside the bound.

Part C IPM

Benefits. Replicate weights give one variance method for gross, net, over- and underpayment rates and for HCC-level or plan-type domains. **Drawbacks.** The national sample is large and stratified, so the CLT should hold, and coverage gains will likely be small. **Implementation.** The recommended approach is a Rao–Wu–Yue rescaled bootstrap within risk strata (resampling $n_h - 1$ per stratum), with percentile and BCa intervals compared against the current method. This would be useful mainly as a cross-check and for domains.

Part D IPM

Benefits. The estimator is a nonlinear simulation (corrected PDEs re-run through benefit phases and reinsurance), so linearization is impractical, and resampling is the natural variance method. CMS describes the estimate as simulation-based and has changed the number of simulation iterations, but the published intervals do not reveal whether the interval is read from those iterations, and if it is, whether the audited rates are held fixed (a plug-in simulation) or redrawn (a parametric bootstrap). This is the first thing a validation should establish, because the companion study finds that the plug-in version does not hold its nominal coverage.

Drawbacks and questions to settle.

- *What is resampled.* Resampling beneficiaries after their payment errors are computed is inexpensive and valid if the simulated error of each beneficiary is fixed. Re-running the benefit simulation inside each replicate is only needed if the simulation itself has stochastic inputs.
- *Replicate count.* The FY 2023 cut in simulation iterations needs a Monte Carlo error target for the 2.5% and 97.5% quantiles (as opposed to only the standard error).
- *Design fidelity.* The FY 2025 switch from cohort to random beneficiary selection

should be mirrored in the resampling scheme (strata, FPC).
- *Benefit redesign.* The Inflation Reduction Act's 2025 Part D redesign (including the $2,000 out-of-pocket cap) changes the benefit simulation for payment years from 2025, reported from FY 2027.

Implementation. The current method should be validated (as opposed to replaced) by building a synthetic beneficiary population from historical PDE and error data, repeating the full sampling-review-simulation pipeline, and measuring the coverage and tail balance of the current interval against BCa and bootstrap-*t*.

PERM

Benefits.

- The overall rate is a composite (FFS + managed care + eligibility − overlap, at federal share), and the national rate pools three cycles. Replicate weights carry every covariance through without hand-derived formulas.
- Beginning in FY 2030, a state eligibility rate above 3% triggers disallowance. With a few hundred cases per state, a one-sided bound near 3% needs calibrated coverage, and whether the statute's "rate" means the point estimate or a bound is a rulemaking question worth raising.

Drawbacks. Systematic selection after sorting by paid amount has no exact variance estimator. In addition, eligibility cases are a second phase attached to sampled payments, difference resolution and appeals change findings after sampling, and the three cycles measure different years.

Implementation. Replicate weights should be generated per state at selection time, treating sorted systematic samples as implicitly stratified (paired adjacent units). The eligibility phase should then be resampled within the sampled payments, and each component and the composite computed per replicate. Because states are sampled independently, national replicates come from summing state replicate totals.

EIPM (APTC)

Benefits. At an error rate under 1%, a 2,000-application sample sits squarely in the rare-error setting where normal upper bounds are liberal and lower bounds are conservative, and BCa helps both tails. SBE measurement beginning in 2027 will produce smaller state samples where this matters more.

Drawbacks. Strata with zero observed errors produce degenerate resamples, and household-level APTC errors are bounded by the monthly APTC and may be all-or-nothing when eligibility fails.

Implementation. The recommended approach is a stratified rescaled bootstrap with BCa and bootstrap-*t*, with strata that have zero or one error flagged and compared against empirical likelihood and a zero-inflated parametric bound.

8. Research agenda: validating comparative accuracy

The strongest design is a design-based Monte Carlo study. Such a study would build realistic finite populations for each program, rerun the program's actual sampling design and estimator thousands of times, and score every interval method against the known truth on the metric that matches how the interval is used. This is the approach Edwards et al. use for FFS audits, extended to bootstrap and audit-finding simulation comparators and to the five CMS designs. The study should be planned and reported in the ADEMP form (aims, data-generating mechanisms, estimands, methods, performance measures) of Morris, White and Crowther (2019), with the Monte Carlo standard error of every coverage estimate reported (Koehler, Brown and Haneuse 2009).

Study pipeline

1. **Build populations.** Unit-level error models are fit to historical review outcomes: HCC-level validation probabilities and hierarchy moves for Part C and RADV; PDE-level documentation failure and pricing discrepancy for Part D; case-level eligibility error for PERM; application-level APTC error for EIPM. These models are imposed on real frame covariates (payments, risk scores, and county rates) to form "plasmode" populations, and stress populations are added on an error-rate grid from 0.01 to 0.95 with right-, left-, and two-spike payment shapes.
2. **Apply the real design.** Frame rules (including the RADV prediction-model frame), strata, sample sizes, and selection mechanics are reproduced (sorted systematic selection for PERM; cohort versus random beneficiary for Part D).
3. **Apply the production estimator.** The production estimator is applied along with plausible alternatives (ratio vs expansion; enrollee-month weighting).
4. **Compute every interval method.** Every interval method (listed below) is computed on each simulated sample.
5. **Score.** Each interval is scored against the population truth, and the process is repeated 10,000 times per scenario cell.
6. **Stress-test.** Results are stress-tested with review measurement error (for example, coder disagreement and overturns on appeal), missing records scored as errors, and frame misspecification.

Comparators

Current analytic interval (z and t, Satterthwaite df where stratified); naive bootstrap percentile; design (rescaled or pseudo-population) bootstrap with percentile, basic, bootstrap-t, and BCa intervals; calibrated (double) bootstrap; plug-in and parametric-bootstrap audit-finding simulation; empirical likelihood; and, for RADV, a hypergeometric or minimum-sum bound as a conservative floor.

Metrics, matched to the interval's use

Use	Primary metric	Secondary metrics
PIIA rate reporting (Part C, Part D, PERM national, EIPM)	Two-sided coverage and balance between the two tails	Width relative to point estimate; bias and RMSE of the rate
Recovery from a lower bound (2012 RADV, PIM-style)	One-sided exceedance: share of samples where the bound exceeds the truth (target ≤ 10% for a 90% bound)	Average recovery as a share of true overpayment; share of bounds above total payment
Extrapolation gate (RADV PY 2019+)	False-extrapolation rate when the true mean change ≤ 0	Recovery error when the gate opens: point estimate minus truth
Threshold test (PERM eligibility vs 3%)	Probability of a state being classified above 3% when its true rate is at or below 3%, and the reverse	Expected disallowance error in dollars

"Viable" should be defined before running the study (for example, one-sided exceedance within nominal + 2 Monte Carlo standard errors in every scenario cell). With 10,000 replications, the Monte Carlo standard error of a 10% exceedance rate is 0.3 percentage points. Viable methods can then be ranked by recovery or width.

Scenario grid

Program	Population source	Sample sizes	Error-rate range to cover
RADV	Historical RADV and Part C IPM enrollee outcomes on real contract frames	35, 50, 100, 200; frames down to 30	Wide: targeted frames may run high
Part C IPM	Part C IPM history	Production national size by stratum	Around 4–8% of payments
Part D IPM	Historical PDE validation results with benefit simulation	Production size	Around 3–5% of payments; documentation failures dominant
PERM	State universes and eligibility review history	Per-state production sizes	1–15% eligibility; dense near 3%
EIPM	FFE application history	~2,000 total; SBE-sized state samples	0.2–2%
CERT	National claims universe by claim type	~37,500 reviewed (FY 2025 counts by claim type)	About 3% (IPPS) to 25% (DMEPOS) by claim type

Computation

A bootstrap inside a Monte Carlo study multiplies costs: 10,000 outer replications × 5,000 resamples is 5×10^7 estimator evaluations per cell. Three practices keep this tractable: precomputing unit-level errors so that each resample is a weighted sum, generating multinomial or rescaled weights in vectorized blocks, and running cells as independent batch jobs over Parquet-staged populations. The warp-speed method of Giacomini, Politis and White (2013) can screen percentile-type intervals with one resample per outer replication before full runs, but it does not cover BCa.

Data and access needs

The study would be strongest with:

- **Methodology plans.** The programs' Sampling and Estimation Methodology Plans (sample sizes, strata, interval algorithms, replicate counts) and the PERM statistical contractor's variance specification.
- **Historical review outcomes,** at the unit level, held under appropriate data use terms. These outcomes would be used to calibrate error models.
- **Agreed metrics.** A primary metric for each program, agreed with the reporting and recovery offices.

Without these, a study can still proceed on synthetic populations calibrated to public aggregates, with illustrative strata where the actual strata are not published. This is the approach of the companion Monte Carlo evaluation, which implements the first stage of this agenda: analytic estimators (expansion, ratio and regression) versus plug-in and parametric-bootstrap audit-finding simulation, at production error rates, with 25 audit years per program. It does not yet cover the survey bootstraps, BCa, bootstrap-*t*, empirical likelihood, the hypergeometric bounds, the one-sided recovery setting, or the error-rate grid above. The companion study found that plug-in audit-finding simulation intervals are invalid: they measure only the variability of reassigning findings across the population and shrink as the population grows relative to the audit. Redrawing the audited rates from their binomial sampling distribution (a parametric bootstrap of the rates) restored coverage for the gross rate. The parametric-bootstrap interval was also narrower than the ratio estimator's, but that width advantage is model-based (it depends on errors being independent of payment size within strata, which was true of the synthetic populations by construction) and is being tested with size-dependent error scenarios before any recommendation rests on it.

9. Conclusions

The methods CMS uses to estimate improper payments are well established in survey sampling, but their accuracy in these settings is largely untested:

- **Ratio-type analytic estimators** dominate production, with normal-theory intervals.
- **Part D** relies on simulation.

The existing evidence points in a consistent direction. Interval accuracy depends on the skew of the errors and the number of nonzero errors in the sample. Normal-theory intervals are conservative when errors are rare and liberal when they are frequent. Second-order bootstrap intervals correct much of the rare-error failure but have barely been tested where errors are frequent and all-or-nothing.

Audit-finding simulation is attractive for nonlinear payment rules. However, its intervals had not been validated before the companion study. Because the plug-in version does not resample the audit, it understates uncertainty in the same way a single imputation does, and the companion study's first-stage results confirm that its intervals are invalid, while the parametric-bootstrap version restores coverage for the gross rate.

The comparative accuracy of these methods should therefore be established empirically before any of them is relied on for new uses. The appropriate design is a Monte Carlo

evaluation against populations with known truth, run under each program's actual design and scored on the metric that matches how the estimate is used. These new uses include smaller audits, state-level PERM disallowances, State-based Exchange measurement, and contract-level recovery.

References (annotated)

Ninety-five sources in seven groups. Linked entries were opened for this review; unlinked classic references are cited from standard bibliographic records and should be checked against the originals before publication.

A. Statute, guidance and program methodology

- **Payment Integrity Information Act of 2019**, P.L. 116-117 (March 2, 2020). Replaced IPIA, IPERA and IPERIA; requires statistically valid improper payment estimates. *Use:* the legal standard any interval method must satisfy.
- **OMB (2021). M-21-19, Appendix C to Circular A-123** (March 5, 2021). Defines improper and unknown payments, the $10 million and 1.5% or $100 million significance thresholds (§II.C), and the Sampling and Estimation Methodology Plan; does not prescribe a confidence level. *Use:* the submission and review route (OMB review, not formal approval) for changing an interval method.
- **CMS (2026). Medicare Program Integrity Manual, Ch. 8** (Rev. 13762). Sets the one-sided 90% lower bound as the default demand, admits coverage cannot always be guaranteed, accepts methods from standard texts and journals, and caps demands at total payment. *Use:* the regulatory door for bootstrap bounds in recovery settings.
- **CMS (2012). Notice of Final Payment Error Calculation Methodology for Part C RADV Contract-Level Audits.** 201 enrollees in three risk-score strata, expansion estimator, two-sided 99% normal bound, FFS adjuster. *Use:* the historical RADV baseline method.
- **CMS (2025). PY 2019 RADV Audit Methods and Instructions** (version 2, June 25, 2025). 35-enrollee SRS from a prediction-model top-decile frame (minimum frame of 30); mean risk-score change with FPC; lower limit of a two-sided 90% interval (z = 1.645) as an extrapolation gate; point-estimate recovery. *Use:* the clearest small-sample target for bootstrap testing.
- **CMS (2026). PY 2021 RADV Audit Methods and Instructions** (May 29). Top-quartile frame; samples of 35–200 by contract size; "lower bound of the 90 percent confidence interval" with $t(n-1)$, sidedness not stated; Taylor linearization for partial-year enrollees; sample-only collection pending appeal. *Use:* the current production design.
- **CMS (2025). RADV Audits Fast Facts** (July). All RADV-eligible contracts audited from PY 2019; 35–200 enrollees per sample from PY 2020. *Use:* scale of the recovery program.
- **CMS (2025). CMS Rolls Out Aggressive Strategy to Enhance and Accelerate Medicare Advantage Audits** (press release, May 21, 2025). All eligible contracts (approximately 550) audited each year; 35 to 200 records per contract. *Use:* source for the number of contracts and the sample-size range.
- **CMS (2023). Medicare and Medicaid Programs; Policy and Technical Changes to the Medicare Advantage, Medicare Prescription Drug Benefit, PACE, Medicaid Fee-For-Service, and Medicaid Managed Care Programs for Years 2020 and 2021 (RADV final rule).** 88 Fed. Reg. 6643–6665 (February 1, 2023), FR Doc. 2023-01942, CMS-4185-F2. Extrapolation from PY 2018, no FFS adjuster, no specific sampling or extrapolation method adopted. *Use:* the rule vacated in 2025.
- **CMS (2025). FY 2025 Medicare Part C Error Rate Findings.** 6.09% rate; error composition; CY 2020 change to the underpayment calculation. Does not state the sample size, strata or interval method. *Use:* Part C targets and error mix.
- **CMS (2023, 2025). FY 2023 and FY 2025 Part D Error Rate Findings.** Report 95% intervals and the FY 2023 changes to simulation iterations, sampling unit and interval calculation; neither states that the interval is read from simulation iterations.

Use: evidence that CMS describes the Part D estimate as simulation-based, and that the interval algorithm is not public.

- **CMS. Medicare Part D IPM Program Background.** Validation findings "applied to a randomly selected 5% sample of the Part D beneficiary population," payment error calculated for that sample and extrapolated to the remaining beneficiaries. *Use:* the only public description of the Part D extrapolation base, and the closest public description of an audit-finding simulation.
- **CMS (2025). PERM Manual** (April). Cycles, universes, components, contractor roles, rolling rate. Does not describe systematic selection or payment strata. *Use:* PERM design reference.
- **CMS (2024). PERM Supporting Statement B**, OMB control no. 0938-0974, ICR 202408-0938-026. Two-phase design, state strata, systematic selection within payment strata, variance formulas, per the supporting statement; the document could not be retrieved for this review and its wording is not independently verified. *Use:* the analytic benchmark a PERM bootstrap must match in the linear case.
- **CMS. PERM Improper Payment Rate Calculation.** How component, state, cycle and national rates combine. *Use:* specifies the composite a replicate method must reproduce.
- **CMS (2025). 2025 Medicare Fee-for-Service Supplemental Improper Payment Data.** CERT sample counts, rates, confidence intervals, overpayments and underpayments by claim type; error categories. *Use:* CERT targets and error mix.
- **CMS (2026). Medicare FFS CERT Methodology.** Claim-type and service-level stratification; generalized estimator reducing to the separate ratio estimator nationally. *Use:* CERT design reference.
- **CMS. Introduction to the CERT Program.** Four claim-type strata, about 37,500 reviewed claims, ±3-point precision target, documentation timelines. *Use:* CERT design facts.
- **CMS (2026). Fiscal Year 2025 Improper Payments Fact Sheet** (January 15, 2026). FY 2025 rates and dollars for Medicare FFS, Part C, Part D, Medicaid, CHIP and APTC; reports no confidence intervals. *Use:* the FY 2025 results table.
- **PaymentAccuracy.gov (2026). HHS CMS Advance Premium Tax Credit.** FY 2022–2025 APTC results: overpayments, underpayments (none since FY 2023), technically improper payment dollars, margins of error and root causes. *Use:* EIPM targets and error mix.
- **CMS (2026). EIPM Improper Payment Findings.** APTC rates by year (0.62% FY 2022, 0.58% FY 2023, 1.01% FY 2024, 0.89% FY 2025); "In FY 2025, 0.43% of 0.89% was made up of Technically Improper Payments." *Use:* source for the technically improper share and the rate series.
- **CMS (2026). EIPM Program Background.** ~2,000 FFE applications; CY 2024 scope. *Use:* APTC design facts.
- **45 CFR §§155.1600–155.1650 (SEIPM).** State Exchange improper payment measurement: applicability beginning January 1, 2027 (§155.1605); three-year cycle (§155.1610); aggregate sample size allocated by expenditures, precision and prior rates (§155.1620); difference resolution and appeals (§155.1630). *Use:* the coming small-sample APTC setting and the source for the 2027 start.
- **P.L. 119-21 §71106 (July 4, 2025).** Disallows federal match on Medicaid eligibility improper payments above a 3% rate from October 1, 2029 and removes the good-faith waiver at 42 CFR 431.1010. Summarized by **Bipartisan Policy Center (April 8, 2026)**, which adds the PERM context. *Use:* turns state PERM eligibility estimates into financial determinations.

B. Oversight, litigation and commentary

- **GAO (2026). *Payment Integrity: Agencies' Estimated Improper Payments Increased to $186 Billion in Fiscal Year 2025*.** GAO-26-108694, April 27, 2026. About $186 billion government-wide; Medicare Advantage $23.7 billion (6.1%). *Use:* context and GAO's standing concern that some estimates are unreliable.
- **GAO (2016). GAO-16-76, *Medicare Advantage: Fundamental Improvements Needed in CMS's Effort to Recover Substantial Amounts of Improper***

Payments (April 8, 2016). Critiques RADV contract selection and timeliness; does not address the extrapolation method. *Use:* background on RADV design weaknesses.

- **HHS OIG (2026). *HHS Met Many Requirements, but Did Not Fully Comply With the Payment Integrity Information Act of 2019 for Fiscal Year 2025*.** OAS-26-17-042, August 2026. Annual review that includes S&EMP evaluation (CCDF, Medicare FFS and the revised Part D plan). *Use:* the reviewer of any CMS method change; source for HHS's FY 2025 non-compliance.
- **Department of Education OIG (2026). *U.S. Department of Education's Compliance with Payment Integrity Information Reporting Requirements for Fiscal Year 2025*.** ED-OIG/A26DC0288, August 4, 2026. Found Pell Grant and Direct Loan estimates unreliable because nonrandom student-level sampling made the confidence intervals inaccurate; a repeat finding since FY 2019 (the FY 2024 report is A25DC0225, May 27, 2025). *Use:* precedent that interval validity is audited.
- ***Humana Inc. v. Becerra*, No. 4:23-cv-00909-O (N.D. Tex. Sept. 25, 2025)**, appeal docketed, ***Humana Inc. v. Kennedy*, No. 25-11293 (5th Cir.)** (argued October 5, 2026; government opening brief, March 21, 2026). Vacated the 2023 RADV rule on procedural grounds under the Administrative Procedure Act. No reporter citation has been confirmed for the district court opinion. *Use:* the legal status of extrapolation.
- **STAT (October 5, 2026).** Report on the Fifth Circuit argument. *Use:* the date of argument.
- **Marshall T (2025). CMS makes significant updates to and expands scope of MA RADV audits.** *Health Law Weekly*, American Health Law Association, September 5, 2025. Compares 2012, PY 2018 and PY 2019 sampling and extrapolation. *Use:* the only public side-by-side of RADV statistical regimes, and the only public description of the PY 2018 design.
- **MACPAC. Payment Error Rate Measurement.** Historical precision targets and state sample-size ranges. *Use:* PERM sample-size context.

C. Audit sampling and extrapolation for payment populations

- **Stringer KW (1963).** Practical aspects of statistical sampling in auditing. *Proceedings of the Business and Economic Statistics Section, American Statistical Association*, 405–411. Origin of the Stringer bound and monetary-unit sampling. *Use:* the source for "Stringer-type bounds."
- **Kaplan RS (1973).** Statistical sampling in auditing with auxiliary information estimators. *Journal of Accounting Research* 11(2):238–258. First to recognize that the zero/nonzero mixture of audit populations undermines difference, ratio and regression estimators. *Use:* origin of the mixture-population problem.
- **Neter J, Loebbecke JK (1975).** *Behavior of Major Statistical Estimators in Sampling Accounting Populations: An Empirical Study.* Auditing Research Monograph No. 2. New York: AICPA. Real accounting populations show normal-theory intervals missing nominal coverage at low error rates. *Use:* the foundational evidence of failure.
- **Fienberg SE, Neter J, Leitch RA (1977).** Estimating the total overstatement error in accounting populations. *Journal of the American Statistical Association* 72(358):295–302. The multinomial bound for overstatement error. *Use:* an alternative to the Stringer bound in the same low-error regime.
- **Cox DR, Snell EJ (1979).** On sampling and the estimation of rare errors. *Biometrika* 66(1):125–132. The canonical model for sampling and estimating rare errors. *Use:* the rare-error setting of EIPM and the sparse error components of every program.
- **Panel on Nonstandard Mixtures of Distributions, National Research Council (1988).** *Statistical Models and Analysis in Auditing: A Study of Statistical Models and Methods for Analyzing Nonstandard Mixtures of Distributions in Auditing.* Washington, DC: National Academy Press (ISBN 0-309-07817-2). Frames audit data as a zero/nonzero mixture and reviews bounds. *Use:* vocabulary for the mixture problem.
- **Biddle GC, Bruton CM, Siegel AF (1990).** Computer-intensive methods in auditing: bootstrap difference and ratio estimation. *Auditing: A Journal of Practice &*

Theory 9(3):92–114. *Use:* early bootstrap ratio estimation in audit.

- **Bickel PJ (1992).** Inference and auditing: the Stringer bound. *International Statistical Review* 60(2):197–209. Theoretical analysis of why the Stringer bound is conservative. *Use:* the theory behind MUS conservatism.
- **Rohrbach KJ (1993).** Variance augmentation to achieve nominal coverage probability in sampling from audit populations. *Auditing: A Journal of Practice & Theory* 12(2):79–97 (end page to verify). Inflates the variance estimate so that the interval reaches nominal coverage in audit populations. *Use:* a direct precursor of the coverage problem this review addresses.
- **Howard RC (1994).** A combined bound for errors in auditing based on Hoeffding's inequality and the bootstrap. *Journal of Business & Economic Statistics* 12(2):437–448. *Use:* a bootstrap bound for audit populations; sits between Biddle et al. (1990) and Helmers (2000).
- **Kvanli AH, Shen YK, Deng LY (1998).** Construction of confidence intervals for the mean of a population containing many zero values. *J. Business & Economic Statistics* 16(3):362–368. *Use:* likelihood comparator for zero-heavy errors.
- **Kvanli AH, Schauer R (n.d.). The bootstrap: what the government auditor should know.** Unpublished manuscript, hosted by the Multistate Tax Commission; undated, no venue. Simulations (2%, 5% and 10% error populations; exponential error sizes; 5,000 samples per cell; 1,000 BCa resamples) and twelve real samples show BCa lower limits closer to nominal and far less conservative than traditional limits when errors are rare, with the gain fading after about 30 expected errors. *Use:* the most direct bootstrap evidence for low-error audits (EIPM, parts of PERM).
- **Helmers R (2000). Inference on rare errors using asymptotic expansions and bootstrap calibration.** *Biometrika* 87(3):689–694. *Use:* calibrated bounds when nonzero errors are few.
- **Edwards D, Ward-Besser G, Lasecki J, Parker B, Wieduwilt K, Wu F, Moorhead P (2003).** The minimum sum method: a distribution-free sampling procedure for Medicare fraud investigations. *Health Services & Outcomes Research Methodology* 4(4):241–263. *Use:* guaranteed-coverage floor.
- **Ignatova I, Edwards D (2008).** Probe samples and the minimum sum method for Medicare fraud investigations. *Health Services & Outcomes Research Methodology* 8:209–221. *Use:* using a probe sample to plan the minimum-sum design.
- **Gilliland D, Feng W (2010).** An adaptation of the minimum sum method. *HSORM* 10(3):154–164. *Use:* refinement of the floor.
- **Edwards D (2011).** On stratified sampling and ratio estimation in Medicare and Medicaid benefit integrity investigations. *Health Services & Outcomes Research Methodology* 11:79–94. DOI 10.1007/s10742-011-0073-7. *Use:* the only published treatment of ratio versus expansion estimation in Medicare audits; the companion study compares exactly these.
- **Gilliland D, Edwards D (2011).** Using randomized confidence limits to balance risk: an application to Medicare investigations. *The American Statistician* 65(3):149–153. *Use:* an alternative treatment of the one-sided recovery bound.
- **Edwards D, Gilliland D, Ward-Besser G, Lasecki J (2015).** Conservative penny sampling. *J. Survey Statistics & Methodology* 3(4):504–523. DOI 10.1093/jssam/smv025. *Use:* conservative bounds for all-or-nothing error.
- **King B, Madansky A (2013).** On sampling design issues when dealing with zeros. *JSSAM* 1(2):144–170. *Use:* warns that careless stratification can void confidence levels.
- **Gunning P, Horgan JM (2004).** A new algorithm for the construction of stratum boundaries in skewed populations. *Survey Methodology* 30(2):159–166. *Use:* the geometric stratification algorithm that Edwards et al. (2023) found most often viable.
- **Horgan JM (2006).** Stratification of skewed populations: a review. *International Statistical Review* 74(1):67–76. *Use:* the review of stratification methods for payment-shaped populations.
- **Edwards D, Liu P, Delage A (2023). Model-based stratification of payment populations in Medicare integrity investigations.** *Survey Methodology* 49(2):607–626 (December 2023). 166 real Medicare samples; all-or-nothing model;

Monte Carlo design testing of stratification algorithms from the CRAN package `stratification` with the authors' `samptest` function; normal bounds fail at high error rates; only the geometric algorithm was viable for more than about half of the test populations. *Use:* the template for this project's validation study.

- **Liu P (2025). Empirical likelihood estimation in Medicare integrity investigations.** Abstract, 65th ISI World Statistics Congress. Preliminary results: EL at n = 30–90 more often maintains 90% confidence than the normal-theory bound on Medicare populations. *Use:* the nearest competing method; check for the full paper.
- **HHS Office of Inspector General. RAT-STATS statistical software** (2010 and 2019 releases; 2010 User Guide). Free software for sample selection and improper payment estimation, the primary statistical tool of OIG's Office of Audit Services and widely used under corporate integrity agreements. *Use:* the de facto standard implementation of the normal-theory bounds this review critiques.

D. Bootstrap theory and interval construction

- **Efron B (1979).** Bootstrap methods: another look at the jackknife. *Annals of Statistics* 7(1):1–26. *Use:* origin.
- **Efron B (1987).** Better bootstrap confidence intervals. *JASA* 82(397):171–185. *Use:* BCa.
- **Hall P (1988).** Theoretical comparison of bootstrap confidence intervals. *Annals of Statistics* 16(3):927–953. *Use:* why bootstrap-*t* and BCa beat percentile under skew.
- **Rubin DB (1981).** The Bayesian bootstrap. *Annals of Statistics* 9(1):130–134. *Use:* the Bayesian alternative to the parametric bootstrap of audited rates; what the parametric-bootstrap audit-finding simulation is not.
- **Rubin DB (1987).** *Multiple Imputation for Nonresponse in Surveys.* New York: Wiley. *Use:* the between-imputation variance component is the exact analogue of the between-audit variance that plug-in audit-finding simulation omits; single imputation corresponds to fixed rates.
- **Efron B, Tibshirani RJ (1993).** *An Introduction to the Bootstrap.* Chapman & Hall. *Use:* the parametric bootstrap (§6.5) and replicate counts for interval endpoints.
- **DiCiccio TJ, Efron B (1996).** Bootstrap confidence intervals. *Statistical Science* 11(3):189–228 (article and discussion). *Use:* review of interval accuracy.
- **Davison AC, Hinkley DV (1997).** *Bootstrap Methods and Their Application.* Cambridge University Press. *Use:* finite-population and stratified resampling chapters.
- **Hesterberg TC (2015).** What teachers should know about the bootstrap: resampling in the undergraduate statistics curriculum. *The American Statistician* 69(4):371–386. DOI 10.1080/00031305.2015.1089789. *Use:* small-sample percentile intervals are too narrow; key for n = 35.
- **Giacomini R, Politis DN, White H (2013).** A warp-speed method for conducting Monte Carlo experiments involving bootstrap estimators. *Econometric Theory* 29(3):567–589. *Use:* cutting simulation cost.
- **Koehler E, Brown E, Haneuse SJ-PA (2009).** On the assessment of Monte Carlo error in simulation-based statistical analyses. *The American Statistician* 63(2):155–162. DOI 10.1198/tast.2009.0030. *Use:* why and how to report the Monte Carlo error of a coverage estimate.
- **Morris TP, White IR, Crowther MJ (2019).** Using simulation studies to evaluate statistical methods. *Statistics in Medicine* 38(11):2074–2102. *Use:* the ADEMP framework for planning and reporting the validation study in Section 8.

E. Finite-population and survey bootstrap

- **Krewski D, Rao JNK (1981).** Inference from stratified samples: properties of the linearization, jackknife and balanced repeated replication methods. *Annals of Statistics* 9(5):1010–1019. *Use:* the asymptotic basis for comparing replication methods under stratified sampling.
- **Bickel PJ, Freedman DA (1984).** Asymptotic normality and the bootstrap in stratified sampling. *Annals of Statistics* 12(2):470–482. *Use:* the first bootstrap under

stratified sampling, and the conditions under which it is consistent.

- **Chao M-T, Lo S-H (1985).** A bootstrap method for finite populations. *Sankhyā A* 47:399–405. *Use:* origin of the without-replacement (pseudo-population) bootstrap lineage.
- **McCarthy PJ, Snowden CB (1985).** The bootstrap and finite population sampling. *Vital and Health Statistics* Series 2, No. 95 (DHHS Pub. No. 85-1369). NCHS. *Use:* with-replacement scaling for stratified designs.
- **Rao JNK, Wu CFJ (1988).** Resampling inference with complex survey data. *JASA* 83(401):231–241. *Use:* the rescaled bootstrap.
- **Kovar JG, Rao JNK, Wu CFJ (1988).** Bootstrap and other methods to measure errors in survey estimates. *Canadian Journal of Statistics* 16(S1):25–45. *Use:* simulation evidence favoring the bootstrap for one-sided intervals.
- **Rao JNK, Wu CFJ, Yue K (1992).** Some recent work on resampling methods for complex surveys. *Survey Methodology* 18(2):209–217. *Use:* replicate-weight form used in software.
- **Sitter RR (1992a).** A resampling procedure for complex survey data. *JASA* 87(419):755–765. *Use:* mirror-match bootstrap for without-replacement designs.
- **Sitter RR (1992b).** Comparing three bootstrap methods for survey data. *Canadian Journal of Statistics* 20(2):135–154. *Use:* simulation comparison of the rescaled, mirror-match and pseudo-population bootstraps.
- **Booth JG, Butler RW, Hall P (1994).** Bootstrap methods for finite populations. *JASA* 89(428):1282–1289. DOI 10.2307/2290991. *Use:* pseudo-population bootstrap for small frames.
- **Shao J, Tu D (1995).** *The Jackknife and Bootstrap.* Springer. *Use:* theory for survey resampling.
- **Valliant R, Dorfman AH, Royall RM (2000).** *Finite Population Sampling and Inference: A Prediction Approach.* New York: Wiley. *Use:* the model-based (prediction) view of estimation from a sample to a finite population; the natural frame for the width gain of audit-finding simulation, which depends on the error model holding in the unaudited population.
- **Wolter KM (2007).** *Introduction to Variance Estimation*, 2nd ed. Springer. *Use:* systematic-sampling variance options relevant to PERM.
- **Preston J (2009).** Rescaled bootstrap for stratified multistage sampling. *Survey Methodology* 35(2):227–234. *Use:* the rescaled bootstrap for designs with the two-phase, multistage structure of PERM.
- **Kolenikov S (2010).** Resampling variance estimation for complex survey data. *Stata Journal* 10(2):165–199 (see also 2008 slides, which attribute the coverage ranking to Krewski and Rao 1981, Rao and Wu 1988, Kovar, Rao and Wu 1988 and Shao 1996 collectively). *Use:* practical method comparison.
- **Antal E, Tillé Y (2011).** A direct bootstrap method for complex sampling designs from a finite population. *JASA* 106(494):534–543. *Use:* direct without-replacement bootstrap.
- **Beaumont J-F, Patak Z (2012).** On the generalized bootstrap for sample surveys with special attention to Poisson sampling. *International Statistical Review* 80(1):127–148. DOI 10.1111/j.1751-5823.2011.00166.x. *Use:* the generalized (weight-based) bootstrap that unifies the survey bootstraps in the Section 4 table.
- **Mashreghi Z, Haziza D, Léger C (2016). A survey of bootstrap methods in finite population sampling.** *Statistics Surveys* 10:1–52. DOI 10.1214/16-SS113. *Use:* the map of pseudo-population, direct and weight-based methods.
- **Lohr SL (2022).** *Sampling: Design and Analysis*, 3rd ed. Boca Raton: Chapman & Hall/CRC. *Use:* the current textbook treatment of the stratified expansion and ratio estimators and their variances.

F. Zero-inflated and skewed means; health-cost analogues

- **Cochran WG (1977).** *Sampling Techniques*, 3rd ed. Wiley (pp. 39–44 for the effect of skewness on the normal approximation). *Use:* the skew-direction rule for bound conservatism.
- **Chen J, Chen SY, Rao JNK (2003).** Empirical likelihood confidence intervals for the

mean of a population containing many zero values. *Canadian Journal of Statistics* 31(1):53–68. *Use:* EL comparator.

- **Zhou XH, Tu W (2000).** Confidence intervals for the mean of diagnostic test charge data containing zeros. *Biometrics* 56(4):1118–1125. *Use:* parametric zero-inflated comparator for health charges.
- **Tian L (2005).** Inferences on the mean of zero-inflated lognormal data: the generalized variable approach. *Statistics in Medicine* 24(20):3223–3232. *Use:* small-sample zero-inflated intervals.
- **Barber JA, Thompson SG (2000).** Analysis of cost data in randomized trials: an application of the non-parametric bootstrap. *Statistics in Medicine* 19(23):3219–3236. *Use:* bootstrap for skewed per-person health spending.
- **Li X, Ding P (2017).** General forms of finite population central limit theorems with applications to causal inference. *JASA* 112(520):1759–1769. *Use:* conditions under which the normal bound is justified.

G. Companion study

- **Cistola A (2026).** *Comparative Accuracy of Improper Payment Estimation Methods in Federal Health Programs: A Monte Carlo Evaluation of Analytic Estimators and Bootstrap Simulation.* Manuscript, `manuscripts/` in this repository. *Use:* implements the first stage of the research agenda in Section 8 (analytic estimators versus plug-in and parametric-bootstrap audit-finding simulation at production error rates, 25 audit years per program) on synthetic populations for all five programs. Finds that plug-in intervals are invalid, that a parametric bootstrap of the audited rates restores coverage for the gross rate, and that the width advantage over the ratio estimator is model-based and is being tested with size-dependent error scenarios.